FIITJEE Students From All Programs have bagged 34 in Top 100, 66 in Top 200 and 174 in Top 500 All India Ranks. FIITJEE Performance in JEE (Advanced), 2014:
2521 FIITJEE Students from Classroom / Integrated School Programs & 3579 FIITJEE Students from All Programs have qualified in JEE (Advanced), 2014.

ALL INDIA TEST SERIES

FIITJEE JEE (Advanced), 2015

FULL TEST – IV

Paper 1

Time Allotted: 3 Hours **Maximum Marks: 243**

- Please read the instructions carefully. You are allotted 5 minutes specifically for this purpose.
- You are not allowed to leave the Examination Hall before the end of the test.

INSTRUCTIONS

A. General Instructions

1. Attempt ALL the questions. Answers have to be marked on the OMR sheets.
2. This question paper contains Three Parts.
3. **Part-I** is Physics, **Part-II** is Chemistry and **Part-III** is Mathematics.
4. Each part is further divided into three sections: **Section-A, Section-B & Section-C**
5. Rough spaces are provided for rough work inside the question paper. No additional sheets will be provided for rough work.
6. Blank Papers, clip boards, log tables, slide rule, calculator, cellular phones, pagers and electronic devices, in any form, are not allowed.

B. Filling of OMR Sheet

1. Ensure matching of OMR sheet with the Question paper before you start marking your answers on OMR sheet.
2. On the OMR sheet, darken the appropriate bubble with black pen for each character of your Enrolment No. and write your Name, Test Centre and other details at the designated places.
3. OMR sheet contains alphabets, numerals & special characters for marking answers.

C. Marking Scheme For All Three Parts.

1. **Section – A (01 – 06)** contains 6 multiple choice questions which have only one correct answer. Each question carries **+3 marks** for correct answer and **– 1 mark** for wrong answer.

   Section – A (07 – 09) contains 3 multiple choice questions which have one or more than one correct answer. Each question carries **+4 marks** for correct answer. There is no negative marking.

   Section-A (10 – 14) contains 2 paragraphs. Based upon paragraph, 2 and 3 multiple choice questions have to be answered. Each question has only one correct answer and carries **+3 marks** for correct answer and **– 1 mark** for wrong answer.
2. **Section – B (01 – 02)** contains 2 Matrix Match Type questions containing statements given in 2 columns. Statements in the first column have to be matched with statements in the second column. Each question carries **+8 marks** for all correct answer. For each correct row **+2 marks** will be awarded. There may be one or more than one correct matching. No marks will be given for any wrong matching in any question. There is no negative marking.
3. **Section-C (01 – 05)** contains 5 Numerical based questions with answers as numerical value from **0 to 9** and each question carries **+4 marks** for correct answer. There is no negative marking.

Name of the Candidate	
Enrolment No.	

Useful Data

PHYSICS

Acceleration due to gravity	$g = 10\ m/s^2$
Planck constant	$h = 6.6 \times 10^{-34}$ J-s
Charge of electron	$e = 1.6 \times 10^{-19}$ C
Mass of electron	$m_e = 9.1 \times 10^{-31}$ kg
Permittivity of free space	$\varepsilon_0 = 8.85 \times 10^{-12}\ C^2/N\text{-}m^2$
Density of water	$\rho_{water} = 10^3\ kg/m^3$
Atmospheric pressure	$P_a = 10^5\ N/m^2$
Gas constant	$R = 8.314\ J\ K^{-1}\ mol^{-1}$

CHEMISTRY

Gas Constant	R	=	$8.314\ J\ K^{-1}\ mol^{-1}$
		=	0.0821 Lit atm $K^{-1}\ mol^{-1}$
		=	$1.987 \approx 2$ Cal $K^{-1}\ mol^{-1}$
Avogadro's Number	N_a	=	6.023×10^{23}
Planck's constant	h	=	6.625×10^{-34} J·s
		=	6.625×10^{-27} erg·s
1 Faraday		=	96500 coulomb
1 calorie		=	4.2 joule
1 amu		=	1.66×10^{-27} kg
1 eV		=	1.6×10^{-19} J

Atomic No: H=1, He = 2, Li=3, Be=4, B=5, C=6, N=7, O=8, N=9, Na=11, Mg=12, Si=14, Al=13, P=15, S=16, Cl=17, Ar=18, K =19, Ca=20, Cr=24, Mn=25, Fe=26, Co=27, Ni=28, Cu = 29, Zn=30, As=33, Br=35, Ag=47, Sn=50, I=53, Xe=54, Ba=56, Pb=82, U=92.

Atomic masses: H=1, He=4, Li=7, Be=9, B=11, C=12, N=14, O=16, F=19, Na=23, Mg=24, Al = 27, Si=28, P=31, S=32, Cl=35.5, K=39, Ca=40, Cr=52, Mn=55, Fe=56, Co=59, Ni=58.7, Cu=63.5, Zn=65.4, As=75, Br=80, Ag=108, Sn=118.7, I=127, Xe=131, Ba=137, Pb=207, U=238.

Physics

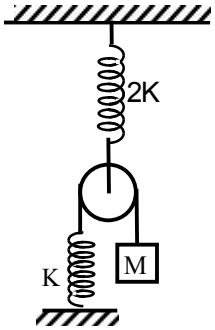
PART – I

SECTION – A

Straight Objective Type

This section contains **6 multiple choice** questions. Each question has four choices (A), (B), (C) and (D) out of which **ONLY ONE** is correct.

1. Find the angular frequency of oscillation of the show system (the pulley is massless).



(A) $\sqrt{\frac{K}{M}}$ (B) $\sqrt{\frac{K}{3M}}$

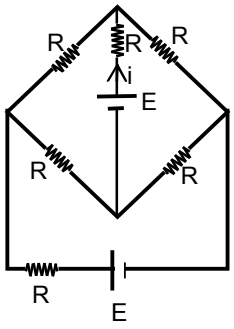
(C) $\sqrt{\frac{K}{2M}}$ (D) $\sqrt{\frac{2K}{3M}}$

2. An object is placed at a distance of 20 cm from an equi-concave lens of focal length 30 cm whose one side is silvered. If a virtual image is formed at 20/3 cm from the lens then what is the radius of curvature of the curved surface?

(A) 30 cm (B) 40 cm

(C) 60 cm (D) 15 cm

3. Find the current i in the circuit.



(A) $i = \frac{E}{2R}$ (B) $i = \frac{E}{R}$

(C) $i = \frac{E}{4R}$ (D) $i = \frac{2E}{R}$

Space for Rough work

4. Four particles each of mass 'm' are located at equal distance from each other. If the separation between any two particles is a then their gravitational potential energy is:

(A) $\frac{-4Gm^2}{a}$ (B) $\frac{-Gm^2}{8a}$

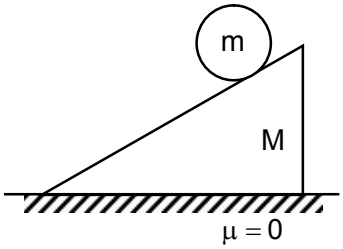
(C) $\frac{-Gm^2}{6a}$ (D) $\frac{-6Gm^2}{a}$

5. A parallel beam of light is incident on a perfectly absorbing small cube of side a. The beam is directed along one of the body diagonals of the cube and has sufficiently large aperture. What is the total radiation force experienced by the cube. I is the intensity of radiation.

(A) $a^2\frac{I}{c}$ (B) $3a^2\frac{I}{c}$

(C) $\sqrt{3}a^2\frac{I}{c}$ (D) $\sqrt{2}a^2\frac{I}{c}$

6. A cylinder is released from top of an inclined plane which itself rests on a smooth horizontal surface. The initial velocity of the cylinder as well as the incline is zero. (Assume no slipping in each case where the incline is rough).



(A) As the cylinder reaches the bottom most point, the inclined plane will acquire a greater speed if it were rough as compared to the case if it were smooth.

(B) As cylinder reaches the bottom-most point of the inclined plane it will acquire a greater speed if it were smooth as compared to the case when it were rough.

(C) The cylinder will have a lower kinetic energy i the case when the inclined plane is rough as compared to the case when it is smooth.

(D) Inclined plane will have a higher kinetic energy in the case when it is rough as compared to the case when it is smooth.

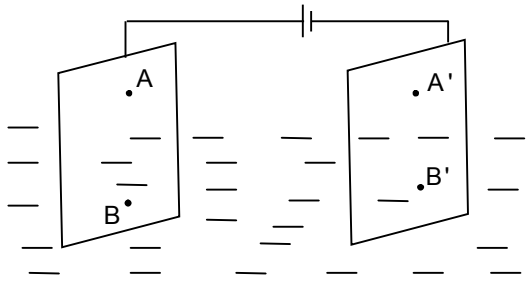
Space for Rough work

Multi Correct Choice Type

This section contains **3 multiple choice** questions. Each question has four choices (A), (B), (C) and (D) out of which **ONE or MORE THAN ONE** is/are correct.

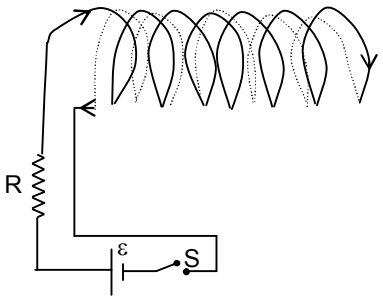
7. Four stationary observers have positions (a, 0), (0, a), (–a, 0) and (0, –a) respectively. A moving source, moving along +x-axis is crossing origin at t = 0. Which of the followings are correct at t = 0?
(A) Wavelength is same for all the observers.
(B) Speed of wave is same for all observers.
(C) Frequency is same for all observers.
(D) Frequency observed is more than emitted by source for three observers.

8. A parallel plate capacitor is dipped in an insulated liquid, vertically so that half of capacitor is in liquid. Plates of capacitor are joined to an ideal battery of emf ε, A, B, A', B' are four points on plate at facing surface as shown. Which of the following is/are correct?



(A) $V_A - V_{B'} = V_A - V_{A'}$
(B) $V_B - V_{B'} = V_B - V_{A'}$
(C) Electric field around A in air = Electric field around B' in liquid.
(D) Electric field around B in liquid = Electric field around B' in liquid.

9. Two identical solenoids each of self inductance L are joined and placed coaxially and overlapping in such a way that they carry equal and opposite currents. This arrangement is joined to a resistance R and battery of emf ε. Switch is closed at t = 0. Identify the correct statements:



(A) Energy stored in the magnetic field increases with time.
(B) Time constant of the circuit is $\frac{2L}{R}$.
(C) Time constant of the circuit is zero.
(D) Power delivered by battery remains constant.

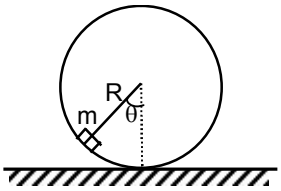
Space for Rough work

Comprehension type (Only One Option Correct)

This section contains **2 paragraphs.** Based upon one of paragraphs **2 multiple choice questions** and based on the other paragraph **3 multiple choice questions** have to be answered. Each of these questions has four choices (A), (B), (C) and (D) out of which **ONLY ONE is correct**.

Paragraph for Question Nos. 10 and 11

A particle of mass m is fixed on a massless ring which rests on a sufficiently rough surface so as to prevent slipping. The system is released from the shown configuration. Then:



10. Immediately after releasing from the shown configuration what is the angular acceleration of the ring?

(A) $\frac{g\sin\theta}{4R\sin^2\left(\frac{\theta}{2}\right)}$ (B) $\frac{g\sin\theta}{2R}$

(C) $\frac{g\sin\theta}{R}$ (D) $\frac{g\sin\theta}{R(1-\cos\theta)}$

11. If α be the λ anticlockwise angular acceleration immediately after release from the shown position, then what is the normal reaction immediately after release:

(A) N = mg – mRαcosθ (B) N = mg

(C) N = m[g – Rαsinθ] (D) None of these

Space for Rough work

Paragraph for Question Nos. 12 and 14

A large cylindrical chamber has a massless piston fitted at the top. An acoustic source is situated at the centre of the bottom of the cylinder, while an observer is situated at the bottom circumference (inside the chamber). The observer is moving directly towards the source with constant velocity and hears a frequency f.

12. If the chamber were to be isobarically heated then the observed frequency will be
(A) same as f (B) higher than f
(C) lower than f (D) data insufficient

13. If the chamber were to be isochorically cooled then the observed frequency will be:
(A) same as f (B) higher than f
(C) lower than f (D) Date insufficient

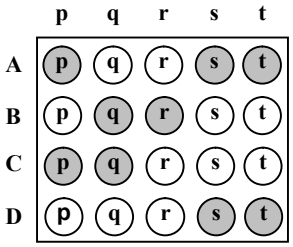
14. If the chamber were to be isothermally heated then, the observed frequency will be:
(A) same as f (B) higher than f
(C) lower than f (D) Data insufficient

Space for Rough work

SECTION - B

Matrix – Match Type

*This section contains 2 questions. Each question contains statements given in two columns, which have to be matched. The statements in **Column I** are labelled A, B, C and D, while the statements in **Column II** are labelled p, q, r, s and t. Any given statement in **Column I** can have correct matching with **ONE OR MORE** statement(s) in **Column II.** The appropriate bubbles corresponding to the answers to these questions have to be darkened as illustrated in the following example:*

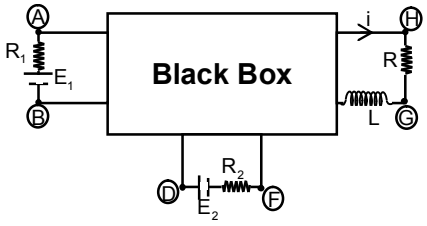


If the correct matches are A – p, s and t; B – q and r; C – p and q; and D – s and t; then the correct darkening of bubbles will look like the following:

1. Initially there are N_0 number of radioactive nuclei of species A while zero nuclei of species B and C. A decays to B with a decay constant λ_1 and B decays to C with a decay λ_2. C is not radioactive. Then match the column:

Column I		Column II	
(A)	$N_B + N_C$	(p)	Continuously increases.
(B)	N_B	(q)	Continuously decreases.
(C)	$N_A + N_C$	(r)	First increases then decreases.
(D)	$N_A + N_B$	(s)	First decreases then increases.
		(t)	Remains constant.

2. A black box having 6 points (A, B, D, F, G, H) contains a network of capacitors, real inductors (with resistance) and resistors. There is no battery inside the black box. E_1 and E_2 are the EMFs of ideal batteries. If we choose $E_1 = E_2 = 1$ volt, then steady state current i is 5A. If we choose $E_1 = 2$ volt and $E_2 = 1$ volt, the steady state current i = 7A. Now match the column:



Column I		Column II	
(A)	$E_1 = 3$ volt, $E_2 = 1$ volt	(p)	i = 10A
(B)	$E_1 = 1$ volt, $E_2 = 2$ volt	(q)	i = 11A
(C)	$E_1 = 1$ volt, $E_2 = 3$ volt	(r)	i = 8A
(D)	$E_1 = 4$ volt, $E_2 = 1$ volt	(s)	i = 9A
		(t)	i = 20 A

Space for Rough work

SECTION – C

(One Integer Value Correct Type)

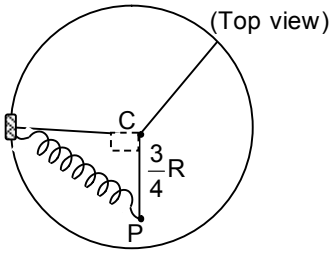
This section contains **5 questions**. Each question, when worked out will result in **one integer** from 0 to 9 (both inclusive).

1. A hydrogen atom in ground state absorbs a photon of energy corresponding to energy difference of first orbit and fourth orbit. After this, electron de-excite from excited state. As a result what maximum number of photons may be obtained?

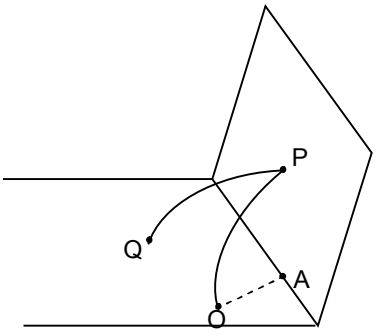
2. A capillary tube is a glass cylinder of radius r = 1 mm. It is dipped vertically in a liquid of surface tension T, $T = 80 \times 10^{-3}$ N/m, and angle of contact $\theta = 60°$, liquid has normal capillary rise in the tube [mass of the tube and its material volume is negligible. Then force required to hold the tube is $P \times 10^{-4}$ N. Calculate P.

3. Two bodies A and B are joined through a conducting rod of high thermal conductivity with insulated cover. A and B are thermally insulated from surrounding. Initial temperature of A and B is 100°C and 20°C respectively. After one hour temperature of A is 80°C and B is 40°C. After further one hour temperature of B is x × 10°C, calculate x?

4. A ring of radius (R = 10 cm) is fixed in a horizontal plane. A sleeve of mass m = 300 gm can move on circumference of ring without friction. Sleeve is joined to a spring of spring constant $K = 200\frac{N}{m}$ and natural length R, other end is fixed at P $\left[CP = \frac{3}{4}R\right]$. Sleeve is released from rest. Calculate the net normal force (in N) exerted by ring on sleeve just after releasing. [$g = 10\ m/s^2$]



5. A ball is thrown form ground towards a vertical wall in such a way that motion plane of ball makes an angle of 30° with normal to wall. Distance of point of projection form wall is $4\sqrt{3}$ m. Collision between ball and wall is elastic. At what distance (in metre) from point of projection ball falls.



Space for Rough work

Chemistry

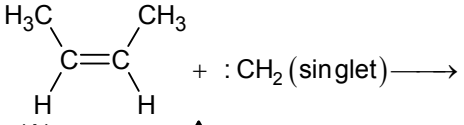
PART – II

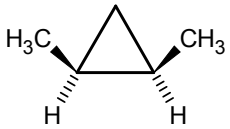
SECTION – A

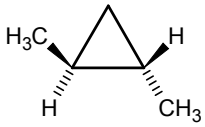
Straight Objective Type

This section contains **6 multiple choice** questions. Each question has four choices (A), (B), (C) and (D) out of which **ONLY ONE** is correct.

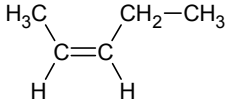
1. The major product formed in the following reaction:

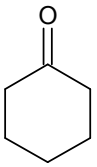


(A) 

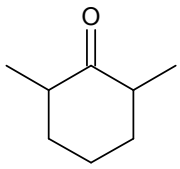
(B) 

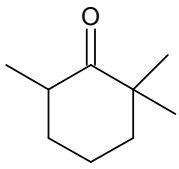
(C) 50 : 50 mixture of above two compounds

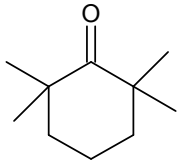
(D) 

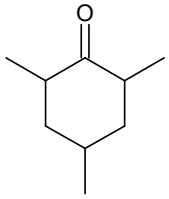
2. 

$\xrightarrow[\text{KH–excess}]{\text{MeI excess}}$ 81% yield, product of the reaction is

(A) 

(B) 

(C) 

(D) 

Space for Rough work

3. Which of the following exhibits aromaticity?
(A) Cyclopentadienyl cation
(B) Cyclopentadienyl radical
(C) Cycloheptatrienyl radical
(D) Cycloheptatrienyl cation

4. $\Delta H < \Delta E$ for the reaction:
(A) $N_2(g) + 3H_2(g) \longrightarrow 3H_2(g)$
(B) $Ag_2O(s) \longrightarrow 2Ag(s) + ½\ O_2(g)$
(C) $C(s) + ½\ O_2(g) \longrightarrow CO(g)$
(D) $C(s) + O_2(g) \longrightarrow CO_2(g)$

5. Which of the following is isoelectronic and isostructural?
$NO_3^-, CO_3^{2-}, ClO_3^-, SO_3$
(A) $NO_3^-,\ CO_3^{2-}$
(B) $SO_3,\ NO_3^-$
(C) ClO_3^-, CO_3^{2-}
(D) CO_3^{2-}, ClO_3^-

6. A particular 100-octane aviation gasoline used 1.00 cm^3 of tetraethyl lead $(C_2H_5)_4Pb$ of density 1.66g/cc, per litre of product. This compound is made as follows.
$4C_2H_5Cl + 4NaPb \longrightarrow (C_2H_5)_4Pb + 4NaCl + 3Pb$. How many gram of ethyl chloride is needed to make enough tetraethyl lead for 1 L of gasoline.
(A) 1.33g
(B) 2.66 g
(C) 9.2g
(D) 0.33g

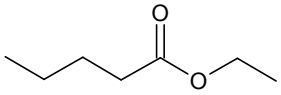
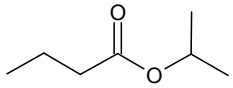
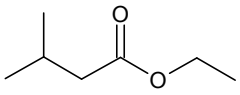
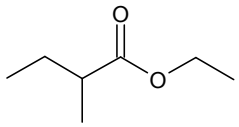
Space for Rough work

Multi Correct Choice Type

This section contains **3 multiple choice** questions. Each question has four choices (A), (B), (C) and (D) out of which **ONE or MORE THAN ONE** is/are correct.

7. The d-orbitals involved in sp^3d^2 or d^2sp^3 hybridisation of the central metal ion are:
   (A) $\mathbf{d_{x^2-y^2}}$ (B) d_{xy}
   (C) d_{yz} (D) $\mathbf{d_{z^2}}$

8. Which of the following statement(s) is/are correct when a soid mixture of NaCl and $K_2Cr_2O_7$ is gently warmed with conc. H_2SO_4?
   (A) A deep red vapour is evolved
   (B) The vapour when passed into NaOH solution gives a yellow soluton of Na_2CrO_4
   (C) Chlorine gas is evolved
   (D) Chromyl chloride is formed

9. Which of the following compound/s shall undergo condensation by reacting with NaOEt/EtOH.
   (A) 
   (B) 
   (C) 
   (D) 

Space for Rough work

Comprehension type (Only One Option Correct)

This section contains **2 paragraphs.** Based upon one of paragraphs **2 multiple choice questions** and based on the other paragraph **3 multiple choice questions** have to be answered. Each of these questions has four choices (A), (B), (C) and (D) out of which **ONLY ONE is correct**.

Paragraph for Question Nos. 10 and 11

Variation of equilibrium constant K with temperature is given by Van't Hoff equation

$$\ln K = \frac{\Delta S_r^o}{R} - \frac{\Delta H_r^o}{RT}$$

From this equation, ΔH_r^o can be calculated if equilibrium constant K_1 and K_2 at two temperature T_1 and T_2 are known.

$$\log\left(\frac{K_2}{K_1}\right) = \frac{\Delta H_r^o}{2.303R}\left[\frac{1}{T_1} - \frac{1}{T_2}\right]$$

10. For an isomerization X (g) $\rightleftharpoons$ Y (g), the temperature dependency of equilibrium constant is given by

$$\ln K = 2 - \frac{1000}{T}$$

the value of ΔS_r^o at 300 K is

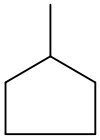
(A) 2R (B) 2/R
(C) 1000 R (D) None of these

11. Select the correct statement
(A) value of K_c always increases with increasing temperature
(B) for exothermic reaction value of K_c increase with decreasing in temperature
(C) for endothermic reaction value of K_c increases with decreasing in temperature
(D) for exothermic reaction slope is $\left(\log K \text{ vs } \frac{1}{T}\right)$ negative

Space for Rough work

Paragraph for Question Nos. 12 to 14

Consider the following reaction



$+ X_2 \xrightarrow{\text{light}}$ mono halogenated product

12. Which halogen will give the best yield of a single mono halogenation products
(A) F_2 (B) Cl_2
(C) Br_2 (D) I_2

13. How many mono halo derivatives are possible (excluding stereo-isomers)?
(A) 3 (B) 4
(C) 5 (D) 6

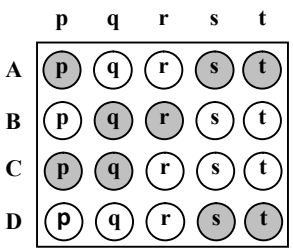
14. Among the following halogens which one is least reactive towards halogenation reaction?
(A) F_2 (B) Cl_2
(C) Br_2 (D) I_2

Space for Rough work

SECTION - B

Matrix – Match Type

*This section contains 2 questions. Each question contains statements given in two columns, which have to be matched. The statements in **Column I** are labelled A, B, C and D, while the statements in **Column II** are labelled p, q, r, s and t. Any given statement in **Column I** can have correct matching with **ONE OR MORE** statement(s) in **Column II.** The appropriate bubbles corresponding to the answers to these questions have to be darkened as illustrated in the following example:*



If the correct matches are A – p, s and t; B – q and r; C – p and q; and D – s and t; then the correct darkening of bubbles will look like the following:

1. Match the following:

Column – I		Column – II	
(A)	Chalcopyrite → copper (pure)	(p)	Froth floatation
(B)	Zinc blende → zinc (pure)	(q)	Hall process
(C)	Bauxite → aluminium (pure)	(r)	Roasting
(D)	Galena → lead (pure)	(s)	Self reduction
		(t)	Basemerisation

2. Match the following:

Column – I		Column – II	
(A)	$K_P > Q_p$	(p)	Non-spontaneous
(B)	$\Delta G^o < RT \log_e Q_p$	(q)	Equilibrium
(C)	$K_P = Q_p$	(r)	Spontaneous and endothermic
(D)	$T > \frac{\Delta H}{\Delta S}$	(s)	Spontaneous
		(t)	Spontaneous and exothermic

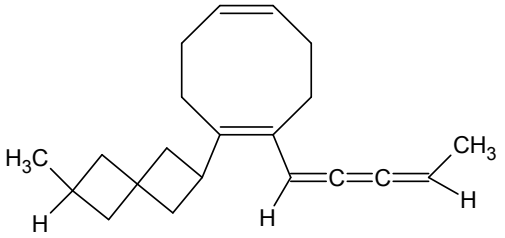
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SECTION – C

(One Integer Value Correct Type)

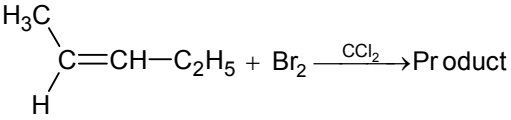
This section contains **5 questions**. Each question, when worked out will result in **one integer** from 0 to 9 (both inclusive).

1. Number of stereo centres in a given molecule?



2. Number of stereoisomers in given molecules [Ma_3b_2C].

3. Number of possible products formed in given reaction



4. The conductivity of 0.001 M Na_2SO_4 solution is 2.6×10^{-4} S cm^{-1} and increases to 7.0×10^{-4} S cm^{-1} and increases to 7×10^{-4} S cm^{-1}, when the solution is saturated with $CaSO_4$. The molar conductivity of Na^+ and Ca^{2+} are 50 and 120 S cm^2 mol^{-1}, respectively. Neglect conductivity of used water the solubility product of $CaSO_4$ is $y \times 10^{-x}$ M^2. What is the value of x.

5. ${}_{90}Th^{234}$ disintegrate to give ${}_{82}Pb^{206}$ as the final product. Total number of β particles emitted out during this process are:

Space for Rough work

Mathematics

PART – III

SECTION – A

Straight Objective Type

This section contains **6 multiple choice** questions. Each question has four choices (A), (B), (C) and (D) out of which **ONLY ONE** is correct.

1. The locus of a point which moves so that the distance between the feet of perpendiculars drawn from it to the lines $3x^2 - 6xy + y^2 = 0$ is $3\sqrt{3}$ is a circle with radius

   (A) $2\sqrt{15}$ (B) 15

   (C) $6\sqrt{5}$ (D) none of these

2. The sum to (n + 1) terms of the series $\cos\alpha + n\cos 2\alpha + \frac{n(n-1)}{2!}\cos 3\alpha + ...$ is

   (A) $2^n \sin^n \frac{\alpha}{2}\cos(2+n)\frac{\alpha}{2}$ (B) $2^n \cos^n \frac{\alpha}{2}\cos(2+n)\frac{\alpha}{2}$

   (C) $2^n \cos(2+n)\frac{\alpha}{2}$ (D) none of these

3. A tangent is drawn to the curve $x^2 + 2x - 4ky + 3 = 0$ at a point whose abscissa is 3. The tangent is perpendicular to the line $x - 2y + 3 = 0$. Then the area bounded by the curve, this tangent, the x-axis and the ordinate $x = -1$, is

   (A) $\frac{108}{47}$ sq. units (B) $\frac{109}{48}$ sq. units

   (C) $\frac{111}{47}$ sq. units (D) $\frac{104}{40}$ sq. units

Space for rough work

4. Given an ellipse $\frac{x^2}{a^2}+\frac{y^2}{b^2}=1$ (a > b). A circle is drawn which passes through the foci and the extremities of the minor axis of this ellipse and let this circle be the auxiliary circle for a second ellipse whose vertices are the foci of the given ellipse. If any tangent to the second ellipse meets its auxiliary circle at two points such that it subtends a right angle at the origin, then the locus of the point of tangency, is (if eccentricity of both the ellipse are same)
(A) $x^2 + 2y^2 = a^2$ (B) $2x^2 + y^2 = a^2$
(C) $2x^2 + y^2 = a^2$ (D) $x^2 + y^2 = 2a^2$

5. The value of $\lim\limits_{x\to\infty}\frac{2(x)^{1/2}+3(x)^{1/3}+4(x)^{1/4}+.....+n(x)^{1/n}}{(2x-3)^{1/2}+(2x-3)^{1/3}+.....+(2x-3)^{1/n}}$ is
(A) 0 (B) 2
(C) $\sqrt{2}$ (D) $\frac{1}{\sqrt{3}}$

6. $\int e^{x^4}\left(x+x^3+2x^5\right)e^{x^2}dx$ is equal to
(A) $\frac{1}{2}xe^{x^2}e^{x^4}+c$ (B) $\frac{1}{2}x^2e^{x^4}+c$
(C) $\frac{1}{2}e^{x^2}e^{x^4}+c$ (D) $\frac{1}{2}x^2e^{x^2}e^{x^4}+c$

Multi Correct Choice Type

This section contains **3 multiple choice** questions. Each question has four choices (A), (B), (C) and (D) out of which **ONE or MORE THAN ONE** is/are correct.

7. If $f(x)=\begin{vmatrix}(1-x)^a & (1+x)^b & 1\\ 1 & (1-x)^a & (1+x)^b\\ (1+x)^b & 1 & (1-x)^a\end{vmatrix}$, then
(A) constant term of f(x) = a + b (B) constant term of f(x) = 0
(C) coefficient of x in f(x) is zero (D) none of these

Space for rough work

8. If $f(a)=\int_0^1 \frac{x^{\cos a}-1}{\ln x}dx$, 'a' being a real number other than an odd multiple of π, then

(A) f(a) = ln(cos a) (B) f(a) = ln(1 + cos a)

(C) f′(a) = $\tan\frac{a}{2}$ (D) f′(a) = $-\tan\frac{a}{2}$

9. If $f(x) = x^3 - x^2 + 100x + 2002$, then

(A) f(1000) > f(1001) (B) $f\left(\frac{1}{2000}\right) > f\left(\frac{1}{2001}\right)$

(C) f(x – 1) > f(x – 2) (D) f(2x – 3) > f(x)

Comprehension type (Only One Option Correct)

This section contains **2 paragraphs.** Based upon one of paragraphs **2 multiple choice questions** and based on the other paragraph **3 multiple choice questions** have to be answered. Each of these questions has four choices (A), (B), (C) and (D) out of which **ONLY ONE is correct**.

Paragraph for Question Nos. 10 and 11

Read the following write up carefully and answer the following questions:
Consider a triangle ABC, where x, y, z are the length of perpendicular drawn from the vertices of the triangle to the opposite sides a, b, c respectively. Let the letters R, r, s, Δ denote the circumradius, in-radius, semi-perimeter and area of the triangle respectively

10. If $\frac{bx}{c}+\frac{cy}{a}+\frac{az}{b}=\frac{a^2+b^2+c^2}{k}$, then the value of k is

(A) R (B) s

(C) 2R (D) $\frac{3}{2}$R

11. If cot A + cot B + cot C = $k\left(\frac{1}{x^2}+\frac{1}{y^2}+\frac{1}{z^2}\right)$, then the value of k is

(A) R^2 (B) rR

(C) Δ (D) $a^2 + b^2 + c^2$

Space for rough work

Paragraph for Question Nos. 12 and 14

Read the following write up carefully and answer the following questions:
If the locus of the circumcentre of variable triangle having sides x-axis, x = 3 and px + qy = 4, where (p, q) lies on the parabola $x^2 = 4ay$ is a curve C, then

12. Co-ordinates of the vertex of this curve C is

(A) $\left(\frac{9}{4}, -\frac{81}{8}a\right)$ (B) $\left(-\frac{9}{4}, -\frac{81}{8}a\right)$

(C) $\left(\frac{9}{4}, -\frac{9}{8}a\right)$ (D) $\left(-\frac{9}{4}, \frac{81}{8}a\right)$

13. The curve C is symmetric about the line

(A) $x = -\frac{9}{4}$ (B) $x = \frac{9}{4}$

(C) $y = -\frac{9}{4}$ (D) $y = \frac{9}{4}$

14. The length of smallest focal chord of this curve C is

(A) $\frac{1}{2a}$ (B) $\frac{1}{4a}$

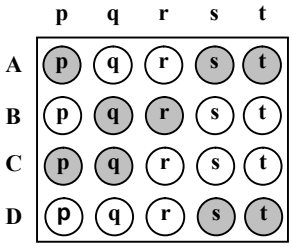
(C) $\frac{1}{12a}$ (D) $\frac{1}{16a}$

Space for rough work

SECTION - B

Matrix – Match Type

*This section contains 2 questions. Each question contains statements given in two columns, which have to be matched. The statements in **Column I** are labelled A, B, C and D, while the statements in **Column II** are labelled p, q, r, s and t. Any given statement in **Column I** can have correct matching with **ONE OR MORE** statement(s) in **Column II.** The appropriate bubbles corresponding to the answers to these questions have to be darkened as illustrated in the following example:*



If the correct matches are A – p, s and t; B – q and r; C – p and q; and D – s and t; then the correct darkening of bubbles will look like the following:

1. Match the following Column-I with Column-II

	Column – I		Column – II
(A)	If A is an idempotent matrix and I is an identity matrix of the same order, then the value of n, such that $(A + I)^n = I + 127A$ is	(p)	9
(B)	If $(I – A)^{-1} = I + A + A^2 + ….. + A^7$, then $A^n = O$ where n is	(q)	10
(C)	If A is a matrix such that $a_{ij} = (i + j) (i – j)$ then A is singular if order of matrix is	(r)	7
(D)	If a non-singular matrix A is symmetric and A^{-1} is also symmetric, then order of A can be	(s)	8
		(t)	6

2. 'n' whole numbers are randomly chosen and multiplied then match the following Column-I with Column-II

	Column–I		Column–II
(A)	The probability that the last digit is 1, 3, 7 or 9 is	(p)	$\frac{8^n + 4^n}{10^n}$
(B)	The probability that the last digit is 2, 4, 6 or 8 is	(q)	$\frac{5^n - 4^n}{10^n}$
(C)	The probability that the last digit is 5 is	(r)	$\frac{4^n}{10^n}$
(D)	The probability that the last digit is zero is	(s)	$\frac{10^n - 8^n - 5^n + 4^n}{10^n}$
		(t)	$\frac{8^n - 4^n}{10^n}$

Space for rough work

SECTION – C
(One Integer Value Correct Type)

This section contains **5 questions**. Each question, when worked out will result in **one integer** from 0 to 9 (both inclusive).

1. Let the circle $(x-1)^2 + (y-2)^2 = 25$ cuts a rectangular hyperbola with transverse axis along $y = x$ at four point A, B, C and D having coordinates (x_i, y_i), (where i = 1, 2, 3, 4) respectively, O being the centre of the hyperbola. If $\ell = x_1 + x_2 + x_3 + x_4$, $m = x_1^2 + x_2^2 + x_3^2 + x_4^2$, $n = y_1^2 + y_2^2 + y_3^2 + y_4^2$, then find value of $n - m - 4\ell$ _____

2. If $x \in [0, 2\pi]$ for which $2\cos x \le \left|\sqrt{1+\sin 2x} - \sqrt{1-\sin 2x}\right| \le \sqrt{2}$ has solution set $x \in \left[\frac{\lambda\pi}{4}, \frac{\mu\pi}{4}\right]$, then $\mu - \lambda$ is _____

3. Given $(1 - 2x + 5x^2 + 10x^3)(1 + x)^n = 1 + a_1x + a_2x^2 + \ldots$ and that $a_1^2 = 2a_2$, then the value of n is _____

4. If $f(x) = a_0 + a_1x + a_2x^2 + \ldots + a_nx^n + \ldots$ and $\frac{f(x)}{1-x} = b_0 + b_1x + b_2x^2 + \ldots + b_nx^n + \ldots$ If $a_0 = 1$ and $b_1 = 3$ and $b_{10} = k^{11} - 1$, then k is _____ (Given that $a_0, a_1, a_2, \ldots$ are in G.P.)

5. For $\tan^{-1}\left\{\frac{\cos 2\alpha \sec 2\beta + \cos 2\beta \sec 2\alpha}{\lambda}\right\} = \tan^{-1}\left\{\tan^2(\alpha+\beta)\tan^2(\alpha-\beta) + \tan^{-1}1\right\}$ the value of λ is _____

Space for rough work